Diffusion Models in Finance: A Survey

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ABSTRACT

Diffusion, score-based, and flow-matching models have rapidly entered quantitative finance as generators for prices, returns, limit order books, financial tables, and structured objects such as volatility surfaces and risk-neutral paths. This survey organizes the emerging literature by the data object modeled by the generator. We review the core modeling families, summarize representative finance applications, and highlight the evaluation problem that currently limits comparison across studies: financial generators must satisfy not only generic distributional fidelity, but also tail behavior, temporal dependence, market-microstructure realism, privacy, and economic constraints. The result is a compact taxonomy and research agenda for diffusion-family models in finance.

KEYWORDS

diffusion models, score-based generative models, flow matching, finance, survey

1 INTRODUCTION

Why diffusion, why now. Generative modeling is useful in finance because many important regimes are rare, private, or expensive to observe: market crashes, stressed liquidity, credit defaults, customer-level fraud, and realistic high-frequency order flow. Denoising diffusion probabilistic models (DDPMs) [18] and score-based generative models [44, 45] offer stable training, strong mode coverage, and flexible conditioning. Those properties make them attractive for synthetic financial data, scenario generation, forecasting, and decision support, where GAN-style instability and mode collapse are especially costly.

Contributions. We make three contributions. First, we introduce a data-first taxonomy that organizes diffusion-family finance papers by the object being modeled: financial time series, limit order books (LOBs), tabular records, and other structured financial objects. Second, we connect those data domains to common applications–generation, prediction and inference, trading and portfolio construction, risk and stress testing, and privacy. Third, we summarize open problems around benchmarks, tail fidelity, market validity, privacy, and non-stationarity.

Scope and taxonomy of this survey. We include learned diffusion, score-based, flow-matching, rectified-flow, and stochastic-interpolant models when they are applied to financial data or provide directly relevant backbones. We exclude classical financial "diffusion" models in the purely SDE/PDE sense unless they contain a learned generative sampler. Figure 1 shows the organizing principle: the data object is the primary axis; application and method family are secondary axes. This differs from generic time-series diffusion surveys [53], tabular-diffusion surveys [6], and broad synthetic-data surveys [29], because finance imposes domain-specific requirements: no-arbitrage and measure consistency, heavy tails, stylized facts, microstructure realism, and privacy/governance.

The rest of the paper follows the same taxonomy. Section 2 gives background. Sections 3–6 review the literature by data domain. Section 7 discusses cross-cutting trends, evaluation, open problems, and directions.

2 BACKGROUND

This section gives the minimum machinery needed to read the survey and draws the bridge to financial modeling.

Denoising diffusion and score-based foundations. DDPMs define a forward process that gradually corrupts data and train a neural network to reverse that corruption [18]. Score-based models learn the score of perturbed data distributions and can be unified through reverse-time SDEs [44, 45]. In finance, this SDE view is more than notation: it connects learned generative sampling to the stochastic-process language already used for prices, rates, volatility, and risk-neutral valuation.

Flow matching, rectified flow, and stochastic interpolants. Flow matching, rectified flow, and stochastic interpolants generalize the same transport idea: learn a path from noise to data [2, 32, 34]. They are increasingly relevant because finance often needs conditional generation, fast sampling, and decision loops where long reverse-diffusion chains are impractical.

Conditioning and guidance. Conditioning is central in finance. Generators are rarely useful if they only sample from an unconditional distribution; users often need scenarios conditioned on regimes, assets, factors, stress levels, option surfaces, or observed market states. Classifier and classifier-free guidance [11, 19] provide generic mechanisms, while finance-specific work adapts conditioning to economic states, cross-sectional relations, and constraints.

Sampling accelerators. Few-step sampling matters for trading, stress testing, and interactive risk systems. DDIM [42], improved diffusion design choices [24, 35], consistency models [43], and flow-based samplers are therefore important backbones even when the final paper is finance-facing rather than method-facing.

Time-series and tabular backbones. General time-series diffusion models such as TimeGrad [36] and CSDI [48], together with tabular models such as TabDDPM [26], provide reusable architectures. Finance papers typically modify these backbones for long memory, heavy tails, cross-sectional dependence, missing data, mixed discrete-continuous fields, or domain constraints.

Diffusion for decision-making and control. Diffusion has also been used as a planner, policy, or trajectory model in reinforcement learning [1, 22, 58]. This matters for finance because portfolio construction, execution, hedging, and market making are decision problems, not merely data generation problems.

3 FINANCIAL TIME SERIES DATA

Financial time series are the largest cluster in the current literature. The modeled data include price paths, return series, volume, factors, firm-characteristic panels, charts, macro variables, and portfolio returns.

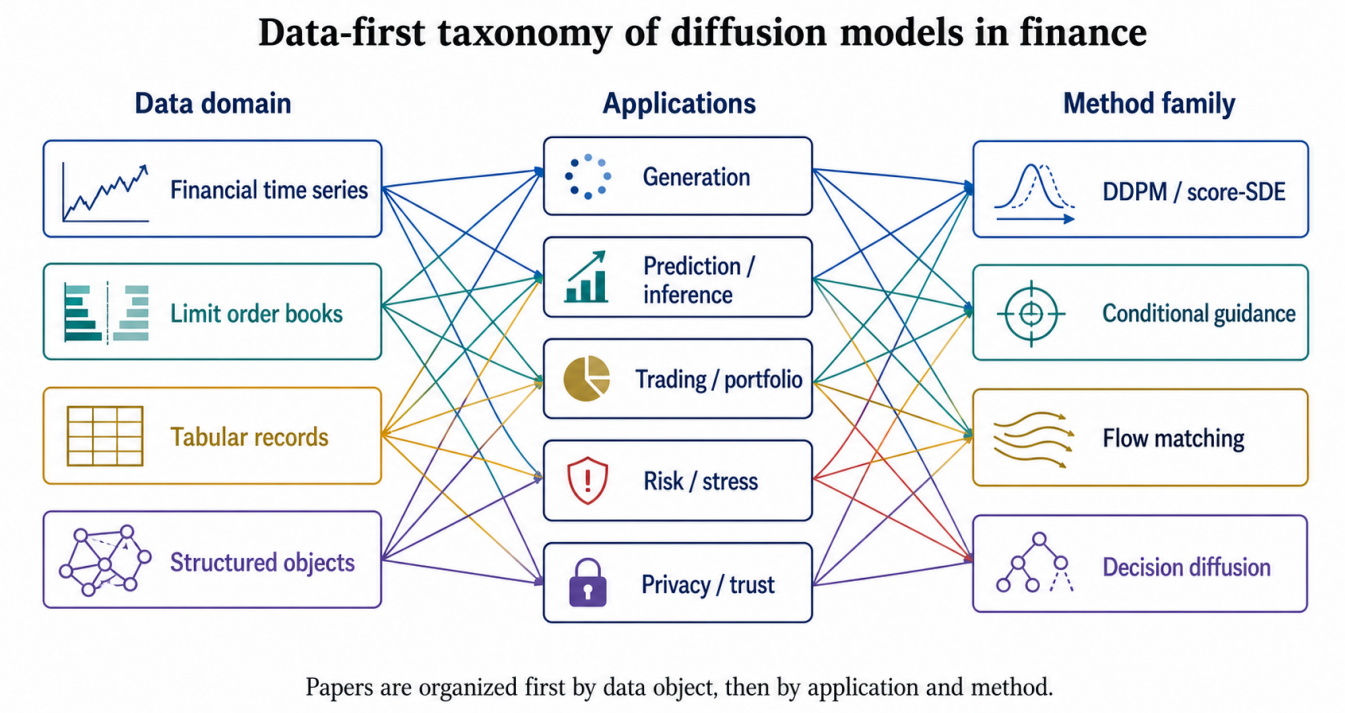


Figure 1: A data-first taxonomy for diffusion-family models in finance. Papers are organized first by the financial data object, then by application and method family.

Table 1: Representative diffusion-family finance papers organized by data domain. The table is illustrative rather than exhaustive; entries are selected to show the taxonomy used in this survey.

Paper	Year	Data domain	Region / Asset / Data	Application	Method family	Code
Beyond Monte Carlo [28]	2024	Time series	US / Stock / Price series	Market simulation	Diffusion	–
A Financial Time Series Denoiser [50]	2024	Time series	US / Stock / Price series	Denoising / trading signal	Conditional diffusion	–
CoFinDiff [47]	2025	Time series	JP / Stock / Price series	Controllable generation	Conditional diffusion	–
Diffusion Factor Models [7]	2025	Time series	US / Stock / Return series	Factor-structured returns	Diffusion	Yes
TRADES [5]	2025	LOB	US / Stock / LOB orderflow	Market simulation	Diffusion	Yes
DiffVolume [51]	2025	LOB	US / Stock / LOB volume	Volume generation	Diffusion	–
DiffLOB [52]	2026	LOB	US / Stock / LOB snapshots	Counterfactual generation	Diffusion	Yes
FinDiff [38]	2023	Tabular	Multi / Credit, Payment, Fund / Tabular	Tabular synthesis	Diffusion	Yes
Imb-FinDiff [40]	2024	Tabular	Multi / Finance / Tabular	Imbalance synthesis	Conditional diffusion	–
DP-FinDiff [41]	2025	Tabular	Multi / Finance / Tabular	Private synthesis	DP diffusion	Yes
IV-surface diffusion [23]	2025	Structured	US / Option / IV surface	Surface forecasting	Diffusion	Yes
Risk-neutral derivative pricing [49]	2026	Structured	Global / Stock / Return series	Derivative pricing	Diffusion	–

3.1 Unconditional Generation

Unconditional generation asks whether a model can synthesize financial series whose marginal distributions, dependence, and stylized facts resemble real markets. Early work generates synthetic asset paths [46] or uses diffusion as a market-dynamics simulator [28]. More recent work adds financial structure: GBM-inspired diffusion processes [25], factor-structured return generation [7], and quality-focused generators such as Fiaingen [37]. The key evaluation question is not simply sample realism, but whether generated paths preserve heavy tails, volatility clustering, cross-sectional dependence, drawdowns, and downstream utility.

3.2 Conditional Generation

Conditional generation uses observed states, regimes, labels, or control signals to steer the generated series. CoFinDiff conditions stock time-series generation on user-specified controls [47]. Stress and inverse-problem papers condition on crisis intensities, compressed observations, or optimization needs [9, 16, 54]. Conditional generation is especially important in finance because users rarely want an average market: they want a plausible market under a specified shock, regime, portfolio state, or observation mask.

3.3 Prediction, Portfolio and Trading

The same models are also used for forecasting, denoising, and decision support. Diffusion-based forecasting and portfolio models target multivariate returns and allocation objectives [8], while DiffSTOCK and DiffsFormer use relational or factor-augmented conditioning for stock prediction [10, 15]. Denoising models refine noisy financial signals before trading [50]. Portfolio work connects generative forecasts to allocation objectives, including mean-variance

and CVaR-aware optimization [3, 13]. Trading-oriented flow models such as FlowHFT treat the generator as part of a policy-learning pipeline [31].

4 LIMIT ORDER BOOKS DATA

LOB data are high-frequency, event-driven, and constrained by market mechanics. They may be represented as snapshots, volume/depth tensors, order-flow sequences, event streams, or byte-level records.

4.1 Generative Market Simulation

LOB simulators aim to produce realistic market dynamics rather than isolated price series. TRADES generates order flow for market simulation [5]; Painting the Market treats LOB states through image-like representations [4]; DiffVolume focuses on volume generation in LOBs [51]; and byte-level or event stream models move closer to raw exchange messages [30, 57]. These works must satisfy stronger tests than low-frequency series: spread, depth, queue dynamics, order signs, price impact, and response to trading flow all matter.

4.2 Prediction

LOB prediction and counterfactual generation ask what the book would look like under an event, intervention, or future market state. DiffLOB targets counterfactual generation of LOB snapshots [52]; diffusion guided meta-agents generate controllable market behavior [20]; and optimal-transport factorization studies market simulation through microstructure regularities [17]. The central difficulty is responsiveness: generated books must react plausibly to actions and shocks rather than merely match unconditional snapshot distributions.

5 TABULAR DATA

Financial tabular data include credit, payment, fund, fraud, customer, and account-level records. These datasets are often mixed-type, imbalanced, privacy-restricted, and constraint-heavy.

5.1 Synthesis and Augmentation

FinDiff introduced finance-specific tabular diffusion across credit, payment, and fund data [38]. Later work extends this direction to entity-based financial tables [33], class imbalance [40], fraud generation [27], and latent-space flow/diffusion augmentation [21]. The common goal is to create tables that improve downstream modeling or enable development when original records cannot be shared.

5.2 Privacy and Trustworthiness

Tabular finance is where privacy and trust become central rather than optional. Federated and differentially private tabular diffusion address the risk of memorizing sensitive records [39, 41]. Privacy-risk studies evaluate utility against membership-inference and reconstruction threats [59]. Trustworthy evaluation must therefore combine utility, privacy leakage, constraint satisfaction, subgroup behavior, and robustness to class imbalance.

6 OTHER STRUCTURED FINANCIAL DATA

Not all financial objects are ordinary series, LOBs, or tables. Some are structured surfaces, matrices, curves, or paths with strong mathematical constraints.

6.1 Volatility, Rates, and Dependence Objects

Implied-volatility surfaces and interest-rate term structures are smooth, arbitrage-sensitive objects indexed by maturity and strike or tenor. Diffusion models have been used for IV-surface forecasting [23] and term-structure generation [12]. Related structured objects include correlation or covariance matrices, where positive semidefiniteness and dependence realism are more important than pixel-like fidelity.

6.2 Derivative Paths, Pricing, and Hedging

Pricing and hedging applications move diffusion models toward risk-neutral generation and stochastic control. Conditional diffusion has been proposed for exotic option valuation [56]; learned samplers have been used for SDE solution paths [14]; and risk-neutral derivative pricing work adapts generative diffusion to valuation under pricing measures [49]. Treasury-futures synthesis similarly shows how conditional time-series diffusion can target structured fixed-income instruments [55].

7 DISCUSSION AND CONCLUSION

7.1 Cross-cutting Methodological Trends

Across data domains, three patterns recur. First, conditioning is becoming the default: regimes, factors, stress levels, observed masks, and market states are what make generated samples useful. Second, finance papers increasingly build structure into the generator or loss, from GBM- and factor-aware time-series models to no-arbitrage and shape constraints in structured objects. Third, diffusion is moving from offline synthesis toward decision pipelines, including portfolio construction, robust reinforcement learning, and trading-aware simulation.

7.2 Evaluation and Benchmarking

Evaluation remains the field's binding constraint. Time-series papers report stylized facts, distributional distances, and downstream forecasting or portfolio metrics; LOB papers need microstructure realism and responsiveness; tabular papers must balance utility and privacy; structured-object papers need validity constraints such as positive semidefiniteness, smoothness, calibration, and no-arbitrage. These metrics are rarely standardized across papers, so claims of superiority are difficult to compare without shared datasets, chronological splits, stress-period tests, and released code.

7.3 Open Problems and Challenges

Important open problems include tail and rare-event fidelity, nonstationarity, economic constraints, computational efficiency, privacy auditing, and causal counterfactuals. Finance also needs benchmarks that test out-of-regime generalization rather than only in-distribution sample quality. For LOBs and trading systems, the next step is interactive evaluation: models should be judged by

how generated markets respond to actions, not only by snapshot similarity.

7.4 Discussion

Diffusion-family models are becoming a general-purpose tool for financial data generation, inference, and decision support. Their promise comes from stable training and flexible conditioning; their risk comes from evaluation gaps and domain constraints that generic generative metrics do not capture. A data-first taxonomy helps clarify what is actually being modeled and what validity checks are required. Future progress will depend less on adding yet another sampler and more on finance-specific benchmarks, transparent reporting, and models that respect market structure.

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